Daniel Foister, Jason Gun, Timyathus Spikes, Tyrese Dickson - DTJT Group
Shehenaz Shaik
CSCI 4350
Sprint #8 - 4/2-4/9

Roles:
- Product Owner: Tyrese Dickson
- Scrum Master: Jason Gun
- Development Team: Daniel Foister, Timyathus Spikes

Time Limit: Two Days (4/2, 4/7, 4/9)
Attendance: All group members were present this week.
Project Github Repo: https://github.com/12061062/StockBuddy
Sprint Goal:
- The goal for this sprint was to improve system security by implementing secure credential handling, input validation, protecting sensitive data, and reinforcing role-based access control. These improvements help prevent unauthorized access, reduce vulnerabilities, and ensure the system handles data securely.

PBIs/User Stories/To-Do Chosen for Sprint:
- **SCRUM-53:** Hash and securely store user credentials instead of using plain text or hardcoded values.
  - Estimation: 4 story points
- **SCRUM-54:** Sanitize input fields to reduce the risk of injection attacks.
  - Estimation: 4 story points
- **SCRUM-55:** Ensure sensitive data is not exposed in logs or error messages.
  - Estimation: 4 story points
- **SCRUM-56:** Reinforce role-based access control across all system features.
  - Estimation: 4 story points
- **==**
- The PBIs/tasks selected for this sprint were based on our team's average velocity of around 16 story points per sprint from previous sprints. These features were prioritized because they improve application security, protect sensitive data, and reduce the risk of vulnerabilities such as unauthorized access and injection attacks.

Sprint Planning
- During sprint planning, our team reviewed the backlog and selected/created four PBIs/tasks that aligned well with StockBuddy's current development stage. The tasks were distributed evenly among our team members, and the acceptance criteria were clarified before beginning implementation.

Total Sprint Duration:
- 4 members x 9 hours/week x 1 Week/Sprint = **36 man-hours available**

Tasks Performed:
- Task Name: Hash and securely store user credentials instead of using plain text or hardcoded values.
- Task Owner: Tyrese Dickson
- Planned Duration: 7 hours
- Actual Duration: 8 hours

- Status: Completed
- Challenges: Ran into some problems with hashing and login functionality.
- Lessons Learned: Secure credential storage is essential for protecting user data.

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- Task Name: Sanitize input fields to reduce the risk of injection attacks.
- Task Owner: Timyathus Spikes
- Planned Duration: 5 hours
- Actual Duration: 6 hours
- Status: Completed
- Challenges: Briefly struggled to validate input fields without breaking certain functions.
- Lessons Learned: None

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- Task Name: Ensure sensitive data is not exposed in logs or error messages.
- Task Owner: Jason Gun
- Planned Duration: 5 hours
- Actual Duration: 6 hours
- Status: Completed
- Challenges: None
- Lessons Learned: None

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- Task Name: Reinforce role-based access control across all system features.
- Task Owner: Daniel Foister
- Planned Duration: 5 hours
- Actual Duration: 6 hours
- Status: Completed
- Challenges: None
- Lessons Learned: None

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Sprint Review:

- Was the Sprint Goal Achieved?:
    - <u>The Sprint Goal was Achieved.</u>
    - Definition of Done (DoD)
        - <u>Our DoD:</u> All work/code maps to the requirements specified in the Jira Board (backlog) and meets the defined acceptance criteria. Each completed item has been fully implemented, tested, and verified to function correctly within the system. Code has been reviewed by team members, integrated into the main project without conflicts, and does not introduce new defects. Any necessary documentation has been updated, and the feature is considered complete, stable, and ready for demonstration or release.
    - Our team successfully implemented secure credential handling, input sanitization, protection of sensitive data, and reinforced role-based access control. These improvements strengthen the system's security and help protect against unauthorized access and common vulnerabilities.

- Focus for the Next Sprint:
    - Emergent Architectures
- PO/Stakeholder Review:
    - The Product Owner reviewed all completed tasks/backlog items:
        - User credentials are securely stored using hashing.
        - Input fields are validated and sanitized to prevent malicious input.
        - Sensitive data is no longer exposed in logs or error messages.
        - Role-based access control is enforced across system failures.
    - Feedback Provided:
        - Consider adding monitoring for suspicious activity.
    - Priority Revisions:
        1. Monitoring and logging improvements moved to the highest priority.
        2. Additional security should be considered.
- Testing
    - During this sprint, we implemented some security-related tests to verify that sensitive data is protected and that system inputs and access controls function correctly. These tests helped ensure that StockBuddy is more secure and resistant to common vulnerabilities.
    - Unit Tests:
        - Test that user credentials are stored securely and not in plain text.
        - Test that input fields reject/sanitize malicious input.
        - Test that sensitive data is not included in logs or error messages.
        - Test that role-based authentication correctly restricts access to protected features.

Sprint Retrospective:

- Changes in the Scrum Process:
    - Again, our team does not believe we need to make any changes to the Scrum Process yet, in the context of this Sprint.
- Start: Incorporating basic security practices during development.
- Stop: Overlooking potential security risks during feature implementation.
- Continue: Frequent push requests/commits, clear communication, and consistent task tracking.
- Other Planned Adjustments/Concerns:
    - No major concerns were identified during this sprint.
- Team Velocity:
    - Sprint 1: 12 story points
    - Sprint 2: 16 story points
    - Sprint 3: 18 story points
    - Sprint 4: 17 story points
    - Sprint 5: 17 story points
    - Sprint 6: 16 story points
    - Sprint 7: 16 story points
    - Sprint 8: 16 story points
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    - Average Velocity: ~16 story points per sprint

Backlog Grooming

- PBIs/Tasks Added/Removed/Reprioritized
  - As a system, I want to trigger specific functions based on user actions so that application processes are handled efficiently using an event-driven approach.
  - As a developer, I want to refactor features into independent functions so that the system can simulate a serverless (FaaS) architecture.
  - As a development team, we want to identify and document how the system could transition to a serverless architecture so that we understand how emerging architectures can be applied to our project.
  - Trigger specific functions based on user actions (checkout | update inventory | log transaction).
  - Refactor a small feature into a standalone function to simulate a serverless (FaaS) approach.
  - Identify parts of the system that could be converted into event-driven functions.
  - Document how the system could transition to a serverless architecture.
- Definition of Ready (DoR)
  - <u>Our DoR:</u> A backlog item is considered ready when it has clearly defined acceptance criteria, estimated story points, and any dependencies have been identified. The task is well-understood by the team, correctly scoped, and small enough to be completed within a single sprint. All required resources, tools, and information are available, allowing the team to begin work without blockers.

Daily Standup #1:

- Duration: 5 Minutes
- Facilitator: Jason Gun

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Tyrese Dickson

1. **What did I do yesterday?**
   I wrapped up working on enhancing the test result output to improve readability and clarity for developers.
2. **What do I plan to do today?**
   Today, I planned out how I am going to hash and securely store user credentials instead of using plain text or hardcoded values.
3. **What are any challenges or impediments that I have experienced?**
   None

Timyathus Spikes

1. **What did I do yesterday?**
   I wrapped up working on implementing a notification mechanism to alert developers when a build or test fails.
2. **What do I plan to do today?**
   Today, I planned out how I am going to sanitize input fields to reduce the risk of injection attacks.
3. **What are any challenges or impediments that I have experienced?**
   None

Jason Gun

1. **What did I do yesterday?**
   I wrapped up working on improving existing test cases based on the analysis of previous test failures.
2. **What do I plan to do today?**
   Today, I planned out how I am going to ensure sensitive data is not exposed in logs or error messages.
3. **What are any challenges or impediments that I have experienced?**
   None

Daniel Foister

1. **What did I do yesterday?**
   I wrapped up working on identifying and resolving recurring defects within the system.
2. **What do I plan to do today?**
   Today, I planned out how I am going to reinforce role-based access control across all system features.
3. **What are any challenges or impediments that I have experienced?**
   None

Daily Standup #2:

- Duration: 5 Minutes
- Facilitator: Jason Gun

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Tyrese Dickson

4. **What did I do yesterday?**
   Yesterday, I planned out how I was going to hash and securely store user credentials instead of using plain text or hardcoded values.
5. **What do I plan to do today?**
   Today, I am going to begin hashing and securely storing user credentials instead of using plain text or hardcoded values, which I planned out during the last session.
6. **What are any challenges or impediments that I have experienced?**
   None

Timyathus Spikes

4. **What did I do yesterday?**
   Yesterday, I planned out how I was going to sanitize input fields to reduce the risk of injection attacks.
5. **What do I plan to do today?**
   Today, I am going to begin sanitizing input fields to reduce the risk of injection attacks, which I planned out during the last session.
6. **What are any challenges or impediments that I have experienced?**
   None

Jason Gun

4. **What did I do yesterday?**
   Yesterday, I planned out how I was going to ensure sensitive data is not exposed in logs or error messages.

5. **What do I plan to do today?**
   Today, I am going to begin ensuring sensitive data is not exposed in logs or error messages, which I planned out during the last session.
6. **What are any challenges or impediments that I have experienced?**
   None

Daniel Foister

4. **What did I do yesterday?**
   Yesterday, I planned out how I was going to reinforce role-based access control across all system features.
5. **What do I plan to do today?**
   Today, I am going to begin to reinforce role-based access control across all system features, which I planned out during the last session.
6. **What are any challenges or impediments that I have experienced?**
   None

Daily Standup #3:

- Duration: 5 Minutes
- Facilitator: Jason Gun

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Tyrese Dickson

7. **What did I do yesterday?**
   I continued working on hashing and securely storing user credentials instead of using plain text or hardcoded values.
8. **What do I plan to do today?**
   Complete my assigned task and plan for Sprint #9.
9. **What are any challenges or impediments that I have experienced?**
   None

Timyathus Spikes

7. **What did I do yesterday?**
   I continued working on sanitizing input fields to reduce the risk of injection attacks.
8. **What do I plan to do today?**
   Complete my assigned task and plan for Sprint #9.
9. **What are any challenges or impediments that I have experienced?**
   None

Jason Gun

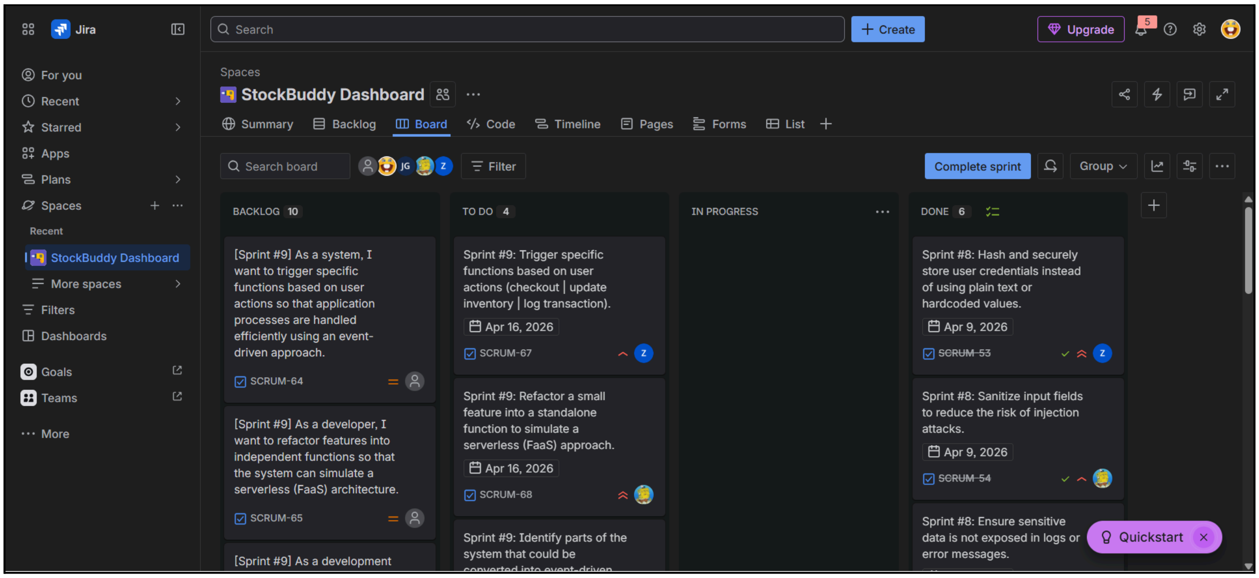
7. **What did I do yesterday?**
   I continued working on ensuring sensitive data is not exposed in logs or error messages.
8. **What do I plan to do today?**
   Complete my assigned task and plan for Sprint #9.
9. **What are any challenges or impediments that I have experienced?**
   None

Daniel Foister

7. **What did I do yesterday?**
   I continued working on reinforcing role-based access control across all system features.

8. **What do I plan to do today?**
   Complete my assigned task and plan for Sprint #9.
9. **What are any challenges or impediments that I have experienced?**
   None

Exported Copy of Scrum Board:



<u>END.</u>